MCSC-201 • 3 CREDITS

Research Methodology

Move from a defensible question to ethical evidence, analysis, interpretation, and publication.

STUDY BLOCKS	FORMAT	SCOPE
10	Definitions Equations Worked examples	Supplied syllabus only

How to use this handbook

For every study block: define the terms, reproduce the relation and its assumptions, solve the example without looking, and explain when the method is unsuitable.

Scope note: This handbook expands only topics named in the supplied syllabus. Confirm examination guidance, prescribed texts, and institutional policy with the responsible instructor.

Course map

1. Foundations and language of research

Meaning, objectives, motivation, and utility, Theory and empiricism, Deductive and inductive theory, Scientific method, Concept, construct, definition, and variable, Research process

2. Problem identification and hypotheses

Research question, Investigation question, Measurement issues, Hypothesis, Good hypothesis qualities, Null and alternative hypothesis, Testing logic and importance

3. Research design

Concept and importance, Features of a good design, Exploratory design, Descriptive design, Experimental design, Independent and dependent variables

4. Qualitative and quantitative approaches

Qualitative research, Quantitative research, Measurement, Causality, Generalization, Replication, Merging approaches

5. Measurement quality and levels

What is measured, Measurement problems, Validity, Reliability, Nominal, Ordinal, Interval, Ratio

6. Sampling

Population, sample, frame, error, size, and non-response, Good sample characteristics, Simple random, Systematic, Stratified random, Multi-stage sampling, Sample-size determination, Practical considerations

7. Data preparation and analysis

Data preparation, Univariate analysis, Frequency tables, Bar and pie charts, Percentages, Bivariate cross-tabulation, Chi-square association test

8. Interpretation and paper writing

Interpretation of data, Research paper layout, Computer science journals, Journal impact factor, When and where to publish, Publishing ethics, Plagiarism and self-plagiarism

9. Research sources and databases

Encyclopedias, Research guides, Handbooks, Academic databases for computer science

10. Research tools and techniques

Effective information search, Zotero and Mendeley, LaTeX and MS Office, Plagiarism-detection software

Study block 01

Foundations and language of research

Meaning, objectives, motivation, and utility	Theory and empiricism
Deductive and inductive theory	Scientific method
Concept, construct, definition, and variable	Research process

Definition and idea

Research is systematic inquiry designed to produce defensible knowledge. Deduction tests consequences derived from theory; induction develops patterns or theory from observations. Concepts become measurable through explicit operational definitions and variables.

```
question -> design -> evidence -> analysis -> interpretation -> communication
```

Worked example

The construct 'system usability' is not directly observed; a study must define indicators, instruments, and scoring before collecting evidence.

> **MUST UNDERSTAND**
> Write one conceptual definition and one operational definition for every key construct.

Study block 02

Problem identification and hypotheses

Research question	Investigation question
Measurement issues	Hypothesis
Good hypothesis qualities	Null and alternative hypothesis
Testing logic and importance	

Definition and idea

A research problem identifies a knowledge gap; a research question makes it answerable. A hypothesis states a testable expectation. The null provides a reference model, while the alternative represents the competing claim.

```
H0: parameter = reference; H1: parameter differs or follows a stated direction
```

Worked example

Question: Does interface A reduce task time compared with B? H0 states no difference in the population; H1 states a difference. The design must define users, tasks, timing, and exclusions.

MUST UNDERSTAND
A useful hypothesis is specific, measurable, falsifiable, and connected to theory.

Study block 03

Research design

Concept and importance	Features of a good design
Exploratory design	Descriptive design
Experimental design	Independent and dependent variables

Definition and idea

Research design is the plan connecting questions to evidence and inference. Exploratory designs clarify poorly understood problems; descriptive designs characterize populations or phenomena; experiments manipulate an independent variable and observe outcomes under controlled comparison.

```
observed effect = treatment effect + bias + random variation
```

Worked example

A randomized experiment estimates the effect of a new interface on completion time. A descriptive survey can estimate current adoption but alone cannot establish the same causal claim.

MUST UNDERSTAND

Align question, sampling, measurement, comparison, analysis, and validity threats.

Study block 04

Qualitative and quantitative approaches

Qualitative research	Quantitative research
Measurement	Causality
Generalization	Replication
Merging approaches	

Definition and idea

Qualitative research investigates meaning and context through rich evidence; quantitative research analyzes measured variables numerically. Mixed approaches integrate them when neither alone answers the full question.

```
triangulation compares evidence from multiple methods or sources
```

Worked example

Interviews can reveal why users abandon a tool, while logs estimate how often and at which step. Integration connects themes with observed behavioral patterns.

MUST UNDERSTAND
Choose methods because they answer the question, not because one label sounds more rigorous.

Study block 05

Measurement quality and levels

What is measured	Measurement problems
Validity	Reliability
Nominal	Ordinal
Interval	Ratio

Definition and idea

Validity concerns whether an inference or instrument measures what it claims. Reliability concerns consistency. Nominal values are categories, ordinal values have order, interval scales have equal differences, and ratio scales also have a meaningful zero.

```
observed score = true component + measurement error
```

Worked example

A 1-to-5 satisfaction item is ordered, but equal spacing between responses is an assumption. Repeating an unreliable instrument cannot repair poor construct validity.

MUST UNDERSTAND
Name the scale level before choosing summaries or statistical tests.

Study block 06

Sampling

Population, sample, frame, error, size, and non-response	Good sample characteristics
Simple random	Systematic
Stratified random	Multi-stage sampling
Sample-size determination	Practical considerations

Definition and idea

A target population is the group of interest; the sampling frame is the operational list used to select units. Probability sampling gives known selection mechanisms. Sample size depends on precision, variability, design, analysis, and practical constraints.

```
SE of a mean decreases approximately as 1/sqrt(n)
```

Worked example

Stratified sampling can ensure representation from small but important groups. Increasing sample size does not remove coverage bias or systematic non-response.

MUST UNDERSTAND
Document target population, frame coverage, selection method, response, and exclusions.

Study block 07

Data preparation and analysis

Data preparation	Univariate analysis
Frequency tables	Bar and pie charts
Percentages	Bivariate cross-tabulation
Chi-square association test	

Definition and idea

Data preparation applies a documented codebook, type checks, missing-value rules, and quality checks. Univariate analysis summarizes one variable. Cross-tabulation and the chi-square test assess association between categorical variables under appropriate assumptions.

```
chi-square = sum((observed-expected)^2/expected)
```

Worked example

For device type by completion status, compute expected counts from row and column totals, inspect whether counts are adequate, calculate chi-square, and report effect context rather than only significance.

MUST UNDERSTAND
Preserve raw data, make transformations reproducible, and distinguish missing from zero.

Study block 08

Interpretation and paper writing

Interpretation of data	Research paper layout
Computer science journals	Journal impact factor
When and where to publish	Publishing ethics
Plagiarism and self-plagiarism	

Definition and idea

Interpretation connects results to the research question, assumptions, uncertainty, limitations, and prior work. A paper typically communicates problem, related context, methods, results, discussion, and references. Venue selection should consider scope, audience, review quality, policies, and fit rather than a metric alone.

```
claim strength must not exceed design and evidence strength
```

Worked example

An observational association should not be written as a causal effect. Reusing substantial prior text or results without transparent citation can be self-plagiarism.

MUST UNDERSTAND
Audit every claim for evidence, uncertainty, attribution, and reproducibility.

Study block 09

Research sources and databases

Encyclopedias	Research guides
Handbooks	Academic databases for computer science

Definition and idea

Reference works orient a topic and supply terminology; research databases help discover primary literature. Search should combine concepts, synonyms, Boolean operators, filters, citation chaining, and a documented inclusion process.

```
search concept blocks: (termA OR synonym) AND (termB OR synonym)
```

Worked example

Begin with a broad scoping search, refine keywords from relevant papers, then trace references backward and citations forward while recording queries and dates.

MUST UNDERSTAND
A reproducible search records database, exact query, date, filters, and screening criteria.

Study block 10

Research tools and techniques

Effective information search	Zotero and Mendeley
LaTeX and MS Office	Plagiarism-detection software

Definition and idea

Reference managers store metadata and attachments and generate citations, but records still require verification. Document tools support structured writing and revision. Similarity tools flag overlap for human review; they do not determine plagiarism by themselves.

```
tool output + researcher judgment + transparent process = defensible workflow
```

Worked example

Import a DOI into a reference manager, verify author and venue metadata, attach the lawful full text, tag the study, cite it through the chosen style, and check the final bibliography.

> **MUST UNDERSTAND**
> Back up the library, verify metadata, preserve notes, and never treat a similarity score as a verdict.

Revision checklist

Use this final pass before an assessment or interview:

- ■ I can define every listed syllabus term in plain language.
- ■ I can reproduce the central equations and state their assumptions.
- ■ I can complete each worked example and create one variation.
- ■ I can compare closely related methods and justify a choice.
- ■ I can name limitations, failure modes, and ethical or operational concerns where relevant.